1. Period $T = 1.33\,s$ Unit 4

$\frac{60s}{T}$

$rpm = \frac{60}{1.33} = 45\,rpm$

$\omega = \frac{2\pi}{1.33} = 4.7$ radians

$= 45\,rpm,\ 4.7$ radians/s

2. $a_c = \omega^2 r$

$r = 0.1\,m$: $\omega = 4.7$ rad/s :

$a_c = (4.7)^2 \times .1 = 2.2\,m/s^2$

$= 2.2\,m/s^2$

3. $100\,a_c = (k\omega)^2 r \Rightarrow k^2 = 100 \rightarrow k = 10$

$\omega \rightarrow 10\omega$

4. $\omega = \frac{200 \times 2\pi}{60} = 20.94$ rad/s

radius $r = .12\,m$

mass $m = .01\,kg$

$F_c = m\omega^2 r = .01 \cdot (20.94)^2 \cdot .12 = .53\,N$

$= .53\,N$

5. $v = \omega r = 30 \times 1.40 = 42\,m/s$

$= 42\,m/s$

Unit 5: Work & Energy

1. $W = \vec{F} \cdot \Delta\vec{x} = (40\hat{i} - 30\hat{j}) \cdot 5\hat{i} = 40 \cdot 5 = 200N$ \ dot pr

F:

$|\vec{F}| = \sqrt{40^2 + (-30)^2} = \sqrt{1600 + 900} = 50\,N$

$\cos\theta = \frac{40}{50} = .8 \rightarrow \theta = \cos^{-1}(.8) = 37°$

$= 200N \cdot m,\ 37°$

2. F=1000N
x=.10m
m=.15 kg

a) $W=F\times X=1000\times.1=$ **100 J**

b) $\Delta KE=$ **100J**

c) $\Delta KE=\frac{1}{2}mv^2\rightarrow100=\frac{1}{2}\cdot.15\cdot v^2$

$v^2=\frac{200}{.15}=1333\rightarrow$ **v=36.5m/s**

3.
m=40kg
μ_k=.05
d=40 m
t=30s

a)
$F_f=.05\cdot40\times9.81=19.62N$
$W=19.62N\times40m=$ **784.8J**

b) $P=\frac{W}{t}$

$P=\frac{784.8J}{30s}=$ **26.16W**

4. P=3.00w
t=1 year=365.24=8760h
9c per kwh

$P_{kW}=\frac{3.00w}{1000}=.003kW$

$E=P_{kW}\times t=.003kW\times8760h=26.28kwh$

26.28kWh·.09USD/1kwh= 2.365USD

= **\$2.33**

Unit 6:

1)

$p_{total} = (20 \cdot 10^{-25}\, kg \cdot 350\, m/s) + (20 \times 10^{-25}\, kg \cdot -350\, m/s)$

$p_{total} = (7 \cdot 10^{-23}\, kg)(m/s) + (-7 \cdot 10^{-23}\, kg)(m/s) = 0$

$v_f = 0\, m/s$

$= 0\, m/s$

2. $m = 20 \cdot 10^{-25}\, kg$

$v = 350\, m/s$

$45°$

$p = m \cdot v = (20 \times 10^{-25}\, kg) \cdot (350\, m/s) = 7 \cdot 10^{-23}\, kg \cdot m/s$

$p_{total} = \sqrt{(7 \cdot 10^{-23})^2 + (7 \cdot 10^{-23})^2 + 2 \cdot (7 \cdot 10^{-23})(7 \cdot 10^{-23}) \cdot \frac{\sqrt{2}}{2}}$

$p_{total} = \sqrt{2 \cdot (7 \cdot 10^{-23})^2 + (7 \cdot 10^{-23})^2 \cdot \sqrt{2}}$

3. $F = 4000\, N$

$t = .200\, s$

$v_i = 2.80\, m/s$

$m = 200\, kg$

a) $J = F \cdot t$

$J = 4000\, N \cdot .200\, s = 800\, N \cdot s$

$= 800\, N$

b) $J = m \cdot (v_f - v_i)$

$v_f = \frac{J}{m} + v_i$

$v_f = \frac{800\, N}{200\, kg} + 2.80\, m/s$

$v_f = 4.00\, m/s + 2.80\, m/s$

$v_f = 6.8\, m/s$

$= 6.80\, m/s$

4. Elastic: 2 objects w/ = mass exchange velocities

15. $m_1v_1 = (m_1+m_2)v_f$

$v_f = \frac{m_1v_1}{m_1+m_2}$

$v_f = \frac{30{,}000 \cdot .850\,m/s}{30{,}000\,kg + 110{,}000\,kg}$

$v_f = \frac{25{,}500}{140{,}000}$

$v_f = .182\,m/s$

$\Delta KE = KE\ initial - KE\ final$

$\Delta KE = 10{,}837.5\,J - 2{,}317\,J$

$\Delta KE = 8{,}520.5\,J$

K.E. lost: 8,520.5J

Unit 7:

$T_1 = 3m\,(40kg \cdot 9.81\,m/s^2)$

$T_1 = 3 \cdot 392.4 = 1177.2N$

Remain motionless!

$T_2 = 2m\,(60kg \cdot 9.81\,m/s^2)$

$T_2 = 2 \cdot 588.6 = 1177.2N$

2. $F_1 + F_2 = W$

$F_1 + F_2 = 196.2N$

$F_2 \cdot 1.50m$

$W \times 1.00m$

$F_2 \cdot 1.50 = W \cdot 1.00$

$F_2 \cdot 1.50 = 196.2 \cdot 1.00$

$F_2 = \frac{196.2}{1.50} = 130.8N$

$F_1 + F_2 = 196.2N$

$F_1 + 130.8 = 196.2$

$F_1 = 65.4N$ (downward F)

$F_2 = 130.8N$ (upward F)

Unit 8:

1. $I = \frac{1}{2}MR^2$ a:

$I = \frac{1}{2} \cdot 1kg(.08m)^2$

$I = \frac{1}{2} \times 1 \cdot .0064$

$= .0032kg$

↓

Moment of Inertia

b) $\tau = I\alpha$

$\tau = .0032 kg \cdot m^2 \cdot .2 rad/s^2$

τ .0064N

↓

Torque

2.

a) $\vec{r} = (-.20\hat{i} + .20\hat{j})m$

$\vec{F} = 12\hat{i}N$

$\vec{\tau} = (r_x F_y - r_y F_x)\hat{k}$

$\vec{\tau} = (-.20 \cdot 0 - .20 \cdot 12)\hat{k}$

$= -2.40\hat{k}N$ Torque

b) $\vec{r} = -.20\hat{i} + .20\hat{j}$

$r_x F_y' - r_y F_x' = 2.40$

$-.20 r_y' - .20 F_x' = 2.40$

$F_y = \frac{2.40}{-.20} = 12N$

$\vec{F} = -12\hat{j}N$ opposite